

Project Central

Project Overview: Probability in the Wild

This quarter, you will follow **one health scenario** through the whole course. Each time we learn a new set of probability tools, you will apply them to your scenario in a checkpoint. At the end of the quarter, you will pull your checkpoints together into a short brief written for a clinician or public health partner.

Every checkpoint follows the same three steps:

1. **Solve it:** Set up the problem with probability notation and solve it analytically (by hand or with R as a calculator).
2. **Simulate it:** Write a simulation in R and check that it agrees with your analytic answer. If the two answers disagree, explain why.
3. **Say it:** Interpret your results in plain language for someone who is not a statistician.

You may work **alone or in pairs**. If you work in a pair, you keep the same partner and scenario for the whole quarter.

Scenario menu

Pick one scenario below, or propose your own in Checkpoint 0. The table shows where each checkpoint could go for each scenario. These are suggestions, not requirements.

Scenario	CP 1: Conditional probability & Bayes	CP 2: Discrete RV	CP 3: Continuous RV & CLT
Cancer screening	PPV/NPV of a screening test	Number of positive screens in a clinic day (Binomial)	Tumor size or biomarker level (Normal)
Emergency department	P(admitted triage level)	Arrivals per hour (Poisson)	Wait time (Exponential)

Scenario	CP 1: Conditional probability & Bayes	CP 2: Discrete RV	CP 3: Continuous RV & CLT
Infectious disease outbreak	$P(\text{infected} \mid \text{positive rapid test})$	Secondary cases from one infected person (Poisson)	Incubation period (Gamma)
Clinical trial recruitment	$P(\text{eligible} \mid \text{screened})$	Patients screened until the first eligible one (Geometric)	Time to enroll n patients (Gamma)
Newborn screening	$P(\text{condition} \mid \text{positive heel-prick test})$	Affected infants per 10,000 births (Binomial / Poisson)	Birth weight (Normal)

i Use real numbers

Wherever you need an input value (prevalence, sensitivity, rate, mean, etc.), find a published estimate and cite it. Good sources include CDC reports, USPSTF evidence reviews, and peer-reviewed articles. You do not need to analyze a dataset. The data in this project come from simulation.

Checkpoints

The checkpoints build, step by step, the work you need for the final project, so we complete the work over the quarter instead of all at the end. **If you receive feedback on something in a checkpoint, you are expected to fix it for the next checkpoint and the final project.**

Checkpoint	Instructions	Points	Rubric	Due Date	Topics
	Checkpoint 0	4		10/18	Choose your scenario + find and cite input values
	Checkpoint 1	12		11/01	Conditional probability + Bayes' rule: when is a positive really positive?

Checkpoint				
Instructions	Points	Rubric	Due Date	Topics
Checkpoint 2	12		11/15	Discrete random variables: counting events
Checkpoint 3	12		11/29	Continuous random variables + the Central Limit Theorem

All checkpoints are submitted as a **rendered Quarto HTML file** (with `code-fold: true`) plus your `.qmd` file.

Final Project: Clinician Brief

The final project is due 12/11.

In the final project, you will revise your three checkpoints based on feedback and combine them into one story about your scenario. You will submit:

1. **Technical appendix:** Your revised Checkpoints 1–3 in a single rendered Quarto file.
2. **Clinician brief:** A 2-page brief (or a 5-minute recorded explanation) for a clinician or public health partner. It should answer: *What does probability tell us about this situation, and what should they keep in mind when making decisions?*

The following links will give you the needed instructions, rubric, and template to complete the final project.

[Final Project Instructions](#)

[Final Project Rubric](#)

[Final Project Template](#)